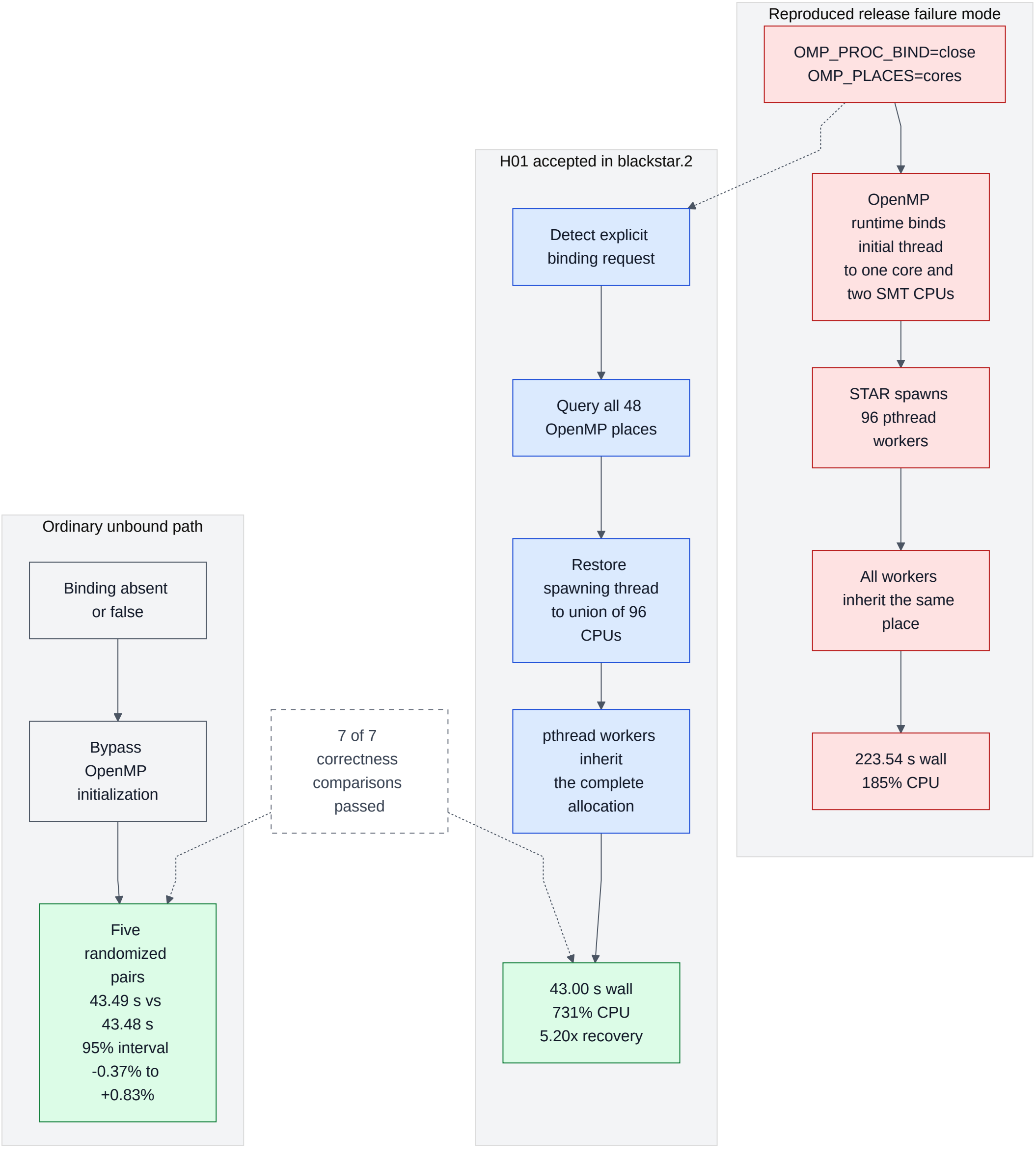
Reproduced release failure mode
OMP_PROC_BIND=close
OMP_PLACES=cores
OpenMP
runtime binds
initial thread
to one core and
two SMT CPUs
STAR spawns
96 pthread
workers
All workers
inherit the same
place
223.54 s wall
185% CPU
H01 accepted in blackstar.2
Detect explicit
binding request
Query all 48
OpenMP places
Restore
spawning thread
to union of 96
CPUs
pthread workers
inherit
the complete
allocation
43.00 s wall
731% CPU
5.20x recovery
7 of 7
correctness
comparisons
passed
Ordinary unbound path
Binding absent
or false
Bypass
OpenMP
initialization
Five
randomized
pairs
43.49 s vs
43.48 s
95% interval
-0.37% to
+0.83%